Uncertainty-Guided Active Learning for Implicit Neural Representations

Piotr Kotłowski 01189039@stud.pw.edu.pl
Stefan Gajda 01189025@stud.pw.edu.pl
Krzysztof Osiński 01189042@stud.pw.edu.pl
Warsaw University of Technology, Poland

Abstract

Reliable uncertainty quantification (UQ) is critical in budget-limited settings, where every observation is costly and a poorly calibrated confidence estimate wastes the next measurement. Prior work on passive INR reconstruction established MC Dropout as the best-calibrated UQ method, but only when the training set is fixed; it is unknown whether this ranking survives *active* learning, where uncertainty estimates choose which coordinates are observed next and small miscalibrations can compound over rounds. We benchmark five UQ methods : MC Dropout, Deep Ensembles, Evidential regression, Last-Layer Laplace, and Gaussian Processes inside a MaxVar active-learning loop across image reconstruction and scientific-discovery tasks. We find that passive calibration rankings do not reliably transfer: rankings depend strongly on the acquisition budget, and the best reconstruction method is not necessarily the best acquisition guide. On Rosenbrock 2D, MC Dropout overtakes even the Gaussian-Process baseline, but only under a large budget; on image reconstruction, Evidential regression gives the best calibration (ECE 0.063) and by far the strongest uncertainty error alignment at low inference cost.

1. Introduction

In many scientific environments, each new observation is expensive. When the measurement budget is fixed, the model itself must decide *where to sample next*, and the quality of that decision rests entirely on its uncertainty estimates. Implicit neural representations (INRs) model a signal as a continuous function from coordinates to values (Sitzmann et al., 2020), which makes them a natural fit for image reconstruction and scientific function approximation under exactly these budget constraints.

Uncertainty quantification (UQ) for INRs has so far been studied almost entirely in the *passive* setting, where the training coordinates are fixed in advance (Duan and Wong, 2026). Vasconcelos et al. (2022) established MC Dropout as the best-calibrated UQ method in that regime. Whether this ranking carries over to *active* learning is unknown. The active setting adds a feedback loop that passive evaluation never exercises: a miscalibrated model can repeatedly acquire uninformative points and compound its own errors over rounds, so a method that is well calibrated on a fixed dataset is not guaranteed to be a good acquisition guide.

Contributions:

- We benchmark five UQ methods: MC Dropout (Gal and Ghahramani, 2016), Deep Ensembles (Lakshminarayanan et al., 2017), Evidential regression (Amini et al., 2020), Last-Layer Laplace (Daxberger et al., 2021), and Gaussian Processes (Rasmussen and Williams, 2006) inside a MaxVar active-learning loop for INRs, across image reconstruction and two scientific-discovery functions.
- We show that passive calibration rankings do not reliably transfer to the active setting: on Rosenbrock 2D, MC Dropout overtakes even the GP baseline, but only under a large acquisition budget (T=20, k=128), while no neural method separates under a small budget.
- We provide a cost-adjusted comparison for image reconstruction and identify Evidential regression as the most practical choice. Best calibration (ECE 0.063) and the strongest uncertainty error alignment (Spearman $\rho = 0.420$ vs. 0.222 for the next-best method) at low inference cost.

2. Methodology

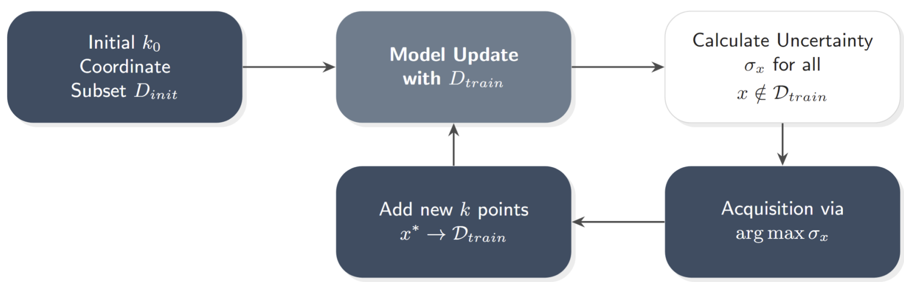

Figure 1: The uncertainty-guided active-learning loop. A SIREN INR is trained on k_0 initial coordinates; each round it predicts values and a per-coordinate uncertainty score, MaxVar selects the k highest-uncertainty coordinates, adds them to the training set, and the model is retrained. This repeats for T rounds.

Problem setup. Given a target signal $f : \mathcal{X} \to \mathbb{R}$ over a coordinate domain $\mathcal{X}$ and a fixed observation budget, we train an INR $\hat{f}_\theta$ to approximate f while choosing which coordinates to reveal. We start from k_0 randomly sampled coordinates. At each of T rounds the model produces, for every candidate coordinate x, a prediction $\hat{f}_\theta(x)$ and an uncertainty score $u(x)$. *MaxVar* acquisition adds the k coordinates with the largest $u(x)$ to the training set, and the model is retrained.

Models. We compare four UQ methods built on a SIREN backbone (Sitzmann et al., 2020) (MC Dropout (Gal and Ghahramani, 2016), Evidential regression (Amini et al., 2020), Deep Ensembles (Lakshminarayanan et al., 2017), and Last-Layer Laplace (Daxberger et al., 2021)) against a Gaussian Process baseline (Rasmussen and Williams, 2006). The methods differ only in how the per-coordinate uncertainty score $u(x)$ is computed: MC Dropout uses the predictive variance over 10 stochastic forward passes; Deep Ensembles use the variance across 5 independently trained members; Evidential regression uses the evidential variance; Last-Layer Laplace uses the last-layer posterior predictive variance; and the GP uses its posterior variance under an RBF kernel. The GP operates over coordinate space rather than learning a neural representation and should be interpreted as a probabilistic baseline rather than an INR.

Training. All neural models use 3 hidden layers of 256 units, learning rate 5×10^{-4}, batch size 32, and one epoch per acquisition round, trained on a single NVIDIA GTX 1080.

Metrics. PSNR and NRMSE measure reconstruction quality; Expected Calibration Error (ECE) measures calibration; and Spearman ρ measures whether high-uncertainty coordinates coincide with large reconstruction errors. which is a direct test of *acquisition* quality, distinct from reconstruction quality. We report means over 3 random seeds.

3. Experiments

3.1 Scientific Discovery Functions

We evaluate reconstruction quality on Rosenbrock 2D and Rastrigin 4D under two budget regimes: small (T=5, k=32) and large (T=20, k=128) with $k_0 = 64$ initial points, averaged over 3 random seeds. Contrasting the two budgets acts as a controlled ablation isolating the effect of acquisition budget from the choice of UQ method.

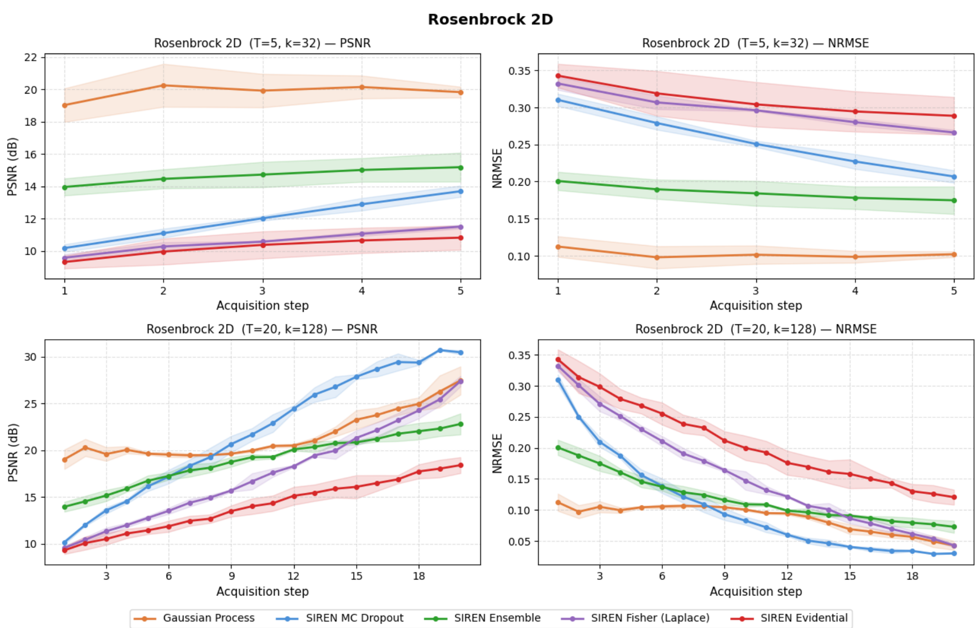

Figure 2: Rosenbrock 2D: PSNR (↑) and NRMSE (↓) under small (top) and large (bottom) budget regimes; shaded regions denote standard deviation over 3 seeds.

On Rosenbrock 2D, the budget is decisive. Under the small budget the GP baseline achieves the strongest reconstruction and the neural UQ methods remain difficult to distinguish. Under the large budget MC Dropout progressively separates from all other methods and eventually surpasses GP by round 20 the only configuration in which a neural method outperforms the probabilistic baseline.

On the harder multimodal Rastrigin 4D (Appendix A, Figure 3) the budget effect persists but is less decisive: neural methods again stagnate under the small budget, while under the large budget Ensemble and MC Dropout converge to competitive performance and Evidential regression lags consistently.

3.2 Image Reconstruction

We evaluate four neural UQ methods on a Kodak image ($k_0 = 10000$, $k = 12000$, $T = 10$). GP is excluded because its $\mathcal{O}(n^3)$ scaling is infeasible at this number of coordinates. FLOPs count multiply and add operations per coordinate at inference.

Table 1: Image reconstruction on a Kodak image (k_0=10000, k=12000, T=10); mean ± std over 3 seeds, best per column in **bold**.

Model	Params	FLOPs	PSNR (dB) ↑	NRMSE ↓	ECE ↓	Spearman ρ ↑
MC Dropout	133,123	2,647,040	21.46 ± 1.60	0.086 ± 0.018	0.080 ± 0.025	0.109 ± 0.036
Ensemble	665,615	1,323,520	$\mathbf{24.13 \pm 1.77}$	$\mathbf{0.064 \pm 0.014}$	0.080 ± 0.023	0.222 ± 0.072
Fisher/Laplace	133,123	**264,704**	20.57 ± 0.89	0.094 ± 0.010	0.561 ± 0.070	0.048 ± 0.069
Evidential	135,436	269,312	22.70 ± 1.88	0.075 ± 0.018	$\mathbf{0.063 \pm 0.018}$	$\mathbf{0.420 \pm 0.046}$

No single method dominates (Table 1): Ensemble wins reconstruction but at five times the parameters, while Evidential gives the best calibration and the closest alignment between uncertainty and error at low cost, making it the better acquisition guide. A ρ of 0.420 is still only moderate, so this alignment remains an open challenge regardless of UQ method.

4. Discussion and Conclusions

What the results mean. Passive calibration rankings do not reliably transfer to active learning. The dominant driver of method rankings is the acquisition budget are T, k, and training epochs per round, often more than the choice of UQ method itself: on Rosenbrock 2D, MC Dropout surpasses even the GP baseline, but only under a sufficiently large budget. The best reconstruction method (Ensemble) is also not the best acquisition guide (Evidential), because reconstruction quality and the alignment between uncertainty and error are distinct objectives.

Where it fails / limitations. Our evidence is scoped. Image reconstruction is evaluated on a single Kodak image, so the ranking there should be read as indicative, not conclusive. We use a single acquisition rule (MaxVar); other rules may reorder methods. Models are trained for one epoch per round, so the under-separation of neural methods under small budgets may be partly an under-training effect. The GP baseline is excluded from image reconstruction due to $\mathcal{O}(n^3)$ scaling, so the neural-vs-probabilistic comparison there is unavailable.

Future work. Larger and more diverse image datasets, acquisition functions beyond MaxVar, and longer training per round (to test whether the gap between uncertainty and error can be closed) are the natural next steps.

Appendix A. Additional Scientific-Discovery Results

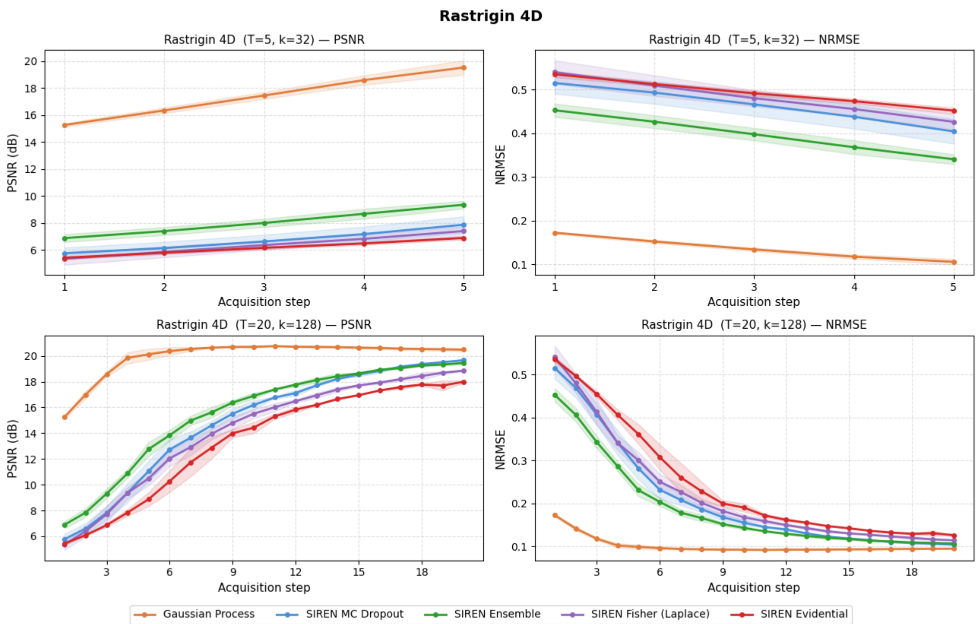

Figure 3: Rastrigin 4D: PSNR (↑) and NRMSE (↓) under small (top) and large (bottom) budget regimes; shaded regions denote standard deviation over 3 seeds. Under the small budget neural methods stagnate; under the large budget Ensemble and MC Dropout converge to competitive performance while Evidential regression lags throughout. This replicates the budget-dependent ranking of Section 3.1 on a harder multimodal target.

References

Alexander Amini, Wilko Schwarting, Ava Soleimany, and Daniela Rus. Deep evidential regression. *Advances in Neural Information Processing Systems (NeurIPS)*, 33:14927–14937, 2020.

Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Randal Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux-effortless bayesian deep learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 34:20089–20103, 2021.

Ruxiao Duan and Alex Wong. Evidential neural radiance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages

28632–28641, 2026.

Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, pages 1050–1059, 2016.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 2017.

Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, and Gordon Wetzstein. Implicit neural representations with periodic activation functions. *Advances in Neural Information Processing Systems (NeurIPS)*, 33:7462–7473, 2020.

Francisca Vasconcelos, Bobby He, Nalini Singh, and Yee Whye Teh. UncertaINR: Uncertainty Quantification of End-to-End Implicit Neural Representations for Computed Tomography. *arXiv preprint arXiv:2202.10847*, 2022.